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Gilbertus Kristo Anugerah Adviandito

Backend Developer · AI Enthusiast

Backend developer and Informatics Engineering student at Universitas Negeri Semarang who builds REST APIs, microservices, and the tests that keep them running. AI enthusiast who trains computer vision models and wires LLMs into backend services. Bangkit Academy graduate with two peer-reviewed papers on image classification.

EDUCATION

Universitas Negeri Semarang (UNNES), Informatics Engineering — 2022 - present
Software engineering, databases, cloud computing, and machine learning.

Bangkit Academy (led by Google, GoTo, and Traveloka), Cloud Computing cohort — Aug - Dec 2023
Backend development, Google Cloud deployment, and ML fundamentals.

SELECTED PROJECTS

Rep Counter · Computer vision — 2026 · demo · source

- Recognizes 22 exercises, counts reps, and flags bad form from a webcam; 0.822 macro-F1 on held-out videos.
- MediaPipe landmarks feed a 1D-CNN exported to ONNX that runs in the browser, so no frames leave the device.

MediaPipe Pose, PyTorch, 1D-CNN, ONNX Runtime Web, Playwright

Pilah · Image classification — 2026 · demo · source

- Waste classifier that tells users which bin an item belongs in; 0.939 macro-F1 on 1,836 test images.
- Fine-tuned MobileNetV3 quantized to a 4.2 MB int8 ONNX model that runs on the phone.

PyTorch, MobileNetV3, ONNX, pytest

Telaah · LLM research automation — 2026 · demo · source

- Turns a research topic into a paper-comparison table from OpenAlex abstracts; code checks every LLM-filled cell against a verbatim quote (98.5% grounding rate on a 16-paper run).
- One core package served as a CLI, an MCP server for agents, and a Streamlit demo.

Python, Gemini API, OpenAlex, MCP, Streamlit, pytest, GitHub Actions

G Pos · Point of sale for food businesses — 2026 · demo · source

- Cashier checkout with discounts, VAT, rounding, and cash, QRIS, or debit payments; owner and cashier roles.
- Logged stock movements, expenses, and profit reports exported to PDF or CSV; 22 PHPUnit test files.

Laravel 12, PHP, MySQL, Alpine.js, Tailwind CSS, Chart.js, Gemini API, PHPUnit

Travel Booking Engine · Distributed systems (in progress) — 2026 · source

- Merges hotel inventory from several suppliers into one search and runs the booking saga to an e-voucher.

Next.js, Express, PostgreSQL, Redis, RabbitMQ, Kafka, Jaeger, k6

PUBLICATIONS

Optimasi Kinerja Klasifikasi Motif Batik pada Arsitektur MobileNetV2 melalui Strategi Fine-Tuning — 2026
G. K. A. Adviandito, M. I. Afif, A. L. Mahendra, M. A. Muslim. *Jurnal Ilmiah Wahana Pendidikan*, 12(6.D), 199-213. Fine-tuning raised accuracy from 94.00% to 99.85%. Link

Implementation of Xception method for Early Detection of Plant Diseases — 2024
G. K. A. Adviandito, A. D. Widiantoro. *SISFORMA*, 11(2), 199-207. 97.63% validation accuracy on 54,305 images.
doi:10.24167/sisforma.v11i2.14741

SKILLS

Backend	Python, TypeScript, Node.js, Express, Flask, Laravel, Prisma, REST APIs, Streamlit
AI & automation	PyTorch, TensorFlow, ONNX Runtime, MediaPipe, OpenCV, scikit-learn, Hugging Face, Tesseract OCR, n8n, Make, Zapier, Claude / Gemini / OpenAI APIs, MCP servers
Data & infrastructure	PostgreSQL, MySQL, MongoDB, Redis, Supabase, RabbitMQ, Kafka, Docker, Google Cloud, Vercel, Prometheus, Grafana, Jaeger
Frontend & testing	React, Next.js, Tailwind CSS, Vitest, Jest, Playwright, pytest, GitHub Actions

ORGANIZATIONS

Chief Executive, Ilkom Social Journey — Nov 2023
Led planning and coordination of the event in Magelang.

Expert Staff, Social Affairs, Hima Ilmu Komputer UNNES — 2023
Designed community programs, including IT training for local residents.

CERTIFICATIONS

Applying Machine Learning with Google Cloud (Dicoding, 2024) · AI Fundamentals (Dicoding, 2024) · Becoming a Google Cloud Engineer (Dicoding, 2024) · Back-End Development with Google Cloud (Dicoding, 2024) · Python for Data Science (HiColleagues, 2024)